



Review Article

Cybersecurity of linear accelerators in radiation oncology beyond ransomware

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ABSTRACT

The rapid digitalization of radiation oncology has enabled transformative advances but also introduced new vulnerabilities for millions of cancer patients. While most reported cyberattacks in healthcare have involved ransomware, the potential compromise of linear accelerators (LINACs) would pose a far greater risk to patient safety. This review examines cybersecurity challenges in radiation oncology with a focus on LINACs. We analyze the anatomy of potential attacks and their implications to raise awareness of stealth threats. We describe LINACs typical network topology and relevant clinical workflow to identify critical assets, vulnerabilities, and attack vectors. We then explore defence mechanisms and real-world best practices, emphasizing that although radiotherapy benefits from mature safety practices and strong clinical safeguards, proactive measures are still needed to anticipate cyber threats and strengthen system resilience. Immediate adoption of a multilayered zero-trust security architecture, supported by robust vendor and regulatory measures, is essential to protect patient safety and maintain clinical continuity in an evolving threat landscape.

Introduction

The annual global cost of cybercrime is estimated in the trillions of dollars, with healthcare representing a major area of concern [1]. The Internet of Medical Things (IoMT) has exposed smart medical devices not only to data breaches but also to life-critical compromises. Cyber incidents affecting healthcare are widely reported in government and industry monitoring, including hundreds of ransomwares and data-breach activities documented by FBI reporting [2]. Among healthcare organizations that experienced ransomware, a 2025 international survey found that 65% cited defensive capacity weaknesses as contributing factors [3]. Notably, 36% of affected healthcare organizations reported paying a ransom, underscoring the continued financial incentives for attackers targeting healthcare environments [3]. Beyond ransomware, healthcare information technology (IT) systems have faced a wide spectrum of threats, from phishing and malware to denial-of-service (DoS)/ distributed denial-of-service (DDoS) and data tampering [4]. These patterns underscore that cyber threats in healthcare are varied,

pervasive, and disruptive.

While most high-profile cybersecurity incidents and defensive efforts in healthcare focus on IT systems, attacks on operational technology (OT) devices, such as infusion pumps, imaging modalities, and LINACs, pose direct risks with higher potential impact on patient safety [5,6]. Research has demonstrated the feasibility of hacking implants like ECGs, blood pressure monitors, and glucose monitors, in some cases potentially delivering lethal doses when linked to pumps [7]. Although this does not indicate that such attacks are common or that LINACs are currently targeted, it illustrates that safety-critical OT systems in healthcare are technically vulnerable. Together with the very low effective enforcement rate for cybercrime ($\approx 0.05\%$ [1]), these factors make healthcare OT assets particularly attractive targets for cybercriminals.

In the past decade, multiple ransomware attacks on cancer centres have been reported worldwide, including in the US, Canada, the UK, and Ireland, some with disastrous outcomes [8–15]. In response, many providers have strengthened IT-focused cybersecurity measures and

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contingency plans to minimize downtime in radiation oncology [6,9,10,12,16–27]. Yet, this raises a critical question: what if LINACs were deliberately hacked to compromise patient safety? Although no such cases have been publicly reported, the possibility of cyber manipulation exists, especially given that fatal non-cyber accidents during radiotherapy have occurred [28,29]. A sophisticated adversary could mimic random errors or plausible complications, avoiding suspicion. While catastrophic scenarios remain improbable, understanding how adversaries might exploit digital workflows is essential for building defences. This review therefore addresses an underexplored gap in the literature, aiming not to suggest imminent danger, but to raise awareness and advocate for stronger safeguards. Unlike prior work focused primarily on ransomware and IT-level disruptions, this review highlights cyberattack pathways targeting LINACs within the broader radiation oncology workflow and introduces a focused threat model with defence recommendations.

Linacs and cybersecurity risks

Why are LINACs potential targets?

Why would criminal organizations invest the resources required for a sophisticated cyberattack on a LINAC? Beyond direct financial incentives, potential motivations include reputational damage, erosion of public trust in healthcare systems, and broader attempts to disrupt critical national infrastructure. History shows that unorthodox and complex cyberattacks have been pursued before, as demonstrated by the 2010 Stuxnet attack on Iran's nuclear program [30] and the 2015–2016

cyberattacks on Ukraine's power grid [31], both of which highlighted how adversaries can disrupt critical infrastructure at scale. Such examples remind us that unlikely scenarios cannot be dismissed outright. This is particularly important given that the lifetime cancer incidence rate is around 40% [32], and more than half of those patients receive radiotherapy [33], meaning millions worldwide depend on reliable access to LINAC treatment. In addition, most of LINACs in operation are manufactured by only two vendors [34]. This creates a degree of uniformity that could be exploited by a sophisticated adversary, since compromising one vendor's ecosystem could potentially scale across many machines. Cybersecurity experts, including Dr. Eric Liederman, have warned of hypothetical extortion scenarios in which attackers could threaten patient safety by shutting down or manipulating critical medical devices, including “*changing the settings of radiation oncology equipment to deliver a deadly amount of radiation to cancer patients*” [35]. Such warnings underscore the importance of considering cyber resilience in safety-critical radiotherapy infrastructure, even in the absence of publicly reported incidents.

LINAC network topology

LINAC is part of a broader networked ecosystem within radiation oncology. As illustrated in Fig. 1, each LINAC is paired with a dedicated Treatment Console Workstation, typically a Windows-based computer, which radiation therapists (RTTs) use to set up and initiate treatment sessions. The treatment console interfaces directly with the LINAC control unit, also referred to as the Treatment Control System (TCS), and serves as the primary user interface for radiation delivery. In addition to

Conceptual Illustration of Key Attack Surfaces

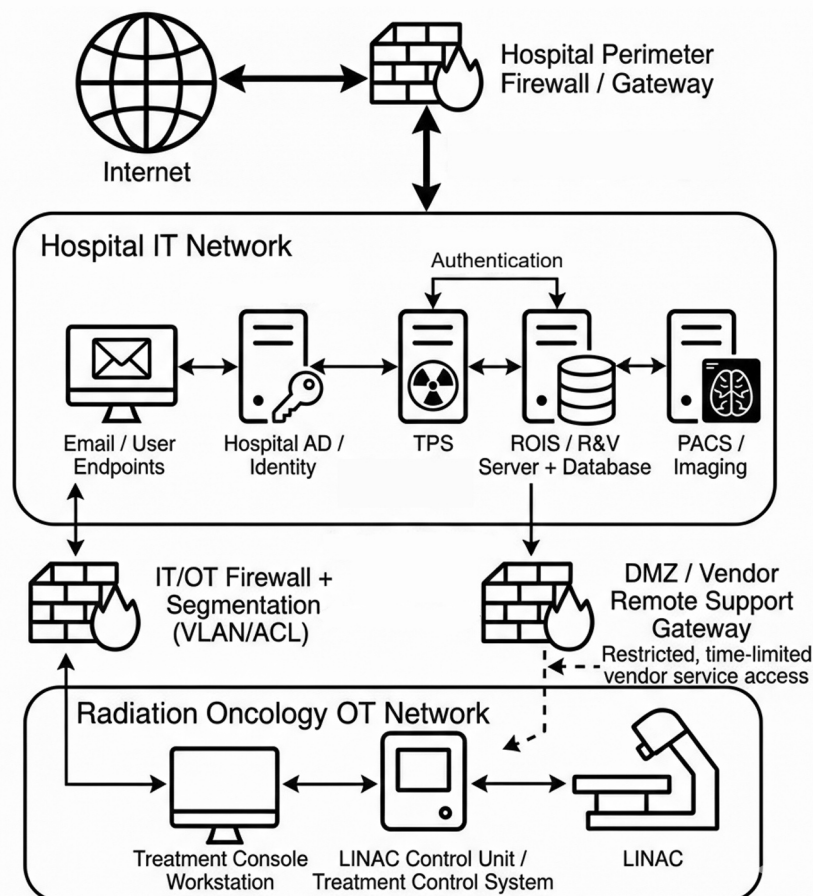


Fig. 1. Simplified IT/OT network topology of a radiation oncology LINAC ecosystem.

application-level software, the LINAC control architecture also includes embedded firmware within the treatment control system, responsible for low-level machine control, safety enforcement, and communication with hardware subsystems. In contemporary clinical deployments, the treatment console workstation is located within the radiation oncology OT network, behind an IT/OT security boundary, together with the LINAC control system. Treatment consoles are typically connected to a radiation oncology information system (ROIS) including its Record & Verify (R&V) component, such as ARIA (Varian) or MOSAIQ (Elekta), which manages scheduling, treatment plan data, and machine parameters. The ROIS/R&V serves as the hub between the treatment planning system (TPS) and the LINAC console. Upstream, TPS platforms such as Eclipse (Varian) or Monaco (Elekta) are used by dosimetrists and physicists to generate treatment plans, which can be exported in DICOM-RT format, depending on the system, to the ROIS/R&V. At treatment time, approved treatment parameters and fraction data are transmitted from the ROIS/R&V system, through the IT/OT firewall and segmentation boundary, to the treatment console workstation for delivery [29].

As shown in Fig. 1, the TPS, ROIS/R&V, and supporting clinical systems typically reside within the hospital IT network, alongside authentication and identity services (e.g. hospital Active Directory (AD), a centralized user and access management service) and imaging infrastructure such as the Picture Archiving and Communication System (PACS, used for storage and distribution of medical images). Fully isolated LINAC networks are uncommon, since imaging, planning, and verification data must flow between hospital systems. Consequently, many centres implement defence-in-depth strategies such as virtual local area network (VLAN, logical network segmentation that restricts which systems can communicate) or firewall-based segmentation (traffic filtering based on predefined security rules), strict access controls, and, in some cases, one-way data diodes (hardware mechanisms that allow data flow in only one direction). Vendors also provide remote support capabilities, which are typically mediated via a dedicated gateway or demilitarized zone (DMZ) and enabled only under controlled conditions [36]. Although such remote support is typically restricted, it can still represent a potential attack surface if access controls are misconfigured, credentials are compromised, or temporary service connections remain active longer than intended. These remote support pathways often rely on authenticated tunnels or service accounts that, if abused, could provide indirect access to systems within the radiation oncology network. Furthermore, because system updates and software patches require vendor validation, clinical staff acceptance, and compliance with medical device regulations, they are slower to apply than on standard IT endpoints, resulting in longer exposure windows [6]. Combined with this interconnectedness, a vulnerability in any single node could provide a foothold into the LINAC control environment.

Workflow touchpoints relevant to cyber risk

While the clinical workflow of radiotherapy is well established, several stages are particularly sensitive to cybersecurity disruption. Here, we highlight key workflow dependencies relevant to cybersecurity risk.

- **Planning-to-delivery data transfer** represents a critical trust boundary. The treatment plan is either transferred internally within an integrated TPS–ROIS environment or exported using DICOM-RT (depending on the system) for subsequent verification and delivery. Any compromise at this stage could propagate downstream without altering the visible clinical workflow.
- **Verification and image-guidance systems**, including kV imaging, CBCT, and SGRT, depend on software-based registration and tolerance thresholds. These systems are designed to detect geometric deviations, but they assume the integrity of both reference data and live acquisition. Attacks that, for example, manipulate reference

images could influence positioning assessment without necessarily producing system alerts.

- **Human oversight** at the console level provides an important safeguard; however, RTTs and physicists primarily validate system outputs rather than underlying data provenance. If compromised data present as internally consistent, routine human checks may not trigger alarms.
- **R&V logs and treatment records** are generally treated as authoritative for QA and incident review. If these records are altered or incomplete, post-treatment detection of anomalous delivery becomes challenging.

This illustrates how radiotherapy safety relies not only on mechanical interlocks but also on the integrity of interconnected digital systems and human oversight. To date, no publicly confirmed cases exist of a LINAC being hacked with the intent to cause patient harm. However, the absence of evidence does not necessarily mean such incidents have never occurred; it may simply mean they have not yet been detected or disclosed.

Threat model

Direct cyberattacks that bypass LINAC hardware interlocks or RTT oversight are highly improbable, due to the presence of multiple independent safety layers, including certified hardware interlocks, real-time dose monitoring, and mandatory human verification steps during treatment setup and delivery. The greater concern is therefore “stealth” manipulations upstream of the gantry, such as altering plan data, exploiting interfaces, or compromising trust in system readouts where errors may go undetected. Our threat model therefore focuses on adversaries exploiting the digital and organisational ecosystem that supports radiotherapy rather than just the LINAC hardware itself. As illustrated in Fig. 2., we consider the following attacker profiles:

- **Opportunistic intruders** who gain access to the hospital network through phishing, unsecured remote access, or compromised devices. Their primary objective is typically operational disruption or financial gain, such as deploying ransomware, with a low likelihood of attempting subtle treatment-related manipulations. Any impact on patient care is therefore more likely to arise indirectly through treatment delays or workflow disruption.
- **Targeted adversaries** with moderate technical skills who can manipulate software systems connected to the LINAC (e.g., ROIS, TPS, or treatment console workstation). These actors present a moderate likelihood of attempting stealth modifications of treatment data or interface spoofing, aiming to remain within tolerance thresholds to avoid detection while influencing clinical decision-making.
- **Advanced persistent threats (APTs)** with significant resources and in some cases, insider knowledge of clinical workflows or vendor ecosystems. While such actors possess the capability to pursue direct manipulation of LINAC firmware or safety interlocks, the likelihood of such attacks remains low, as simpler upstream manipulations of planning, verification, or trust mechanisms are generally more scalable, less detectable, and sufficient to achieve malicious objectives.

Although the likelihood of such attacks is generally low, their potential clinical impact and the difficulty of detection motivate proactive consideration within radiation oncology workflows. Within this scope, the most credible attack surfaces are:

1. **Network and software footholds** in hospital IT or ROIS, enabling tampering with treatment data in transit or at rest.
2. **OIS/TPS manipulation**, where an adversary alters treatment plans, prescription parameters, or associated metadata prior to delivery.

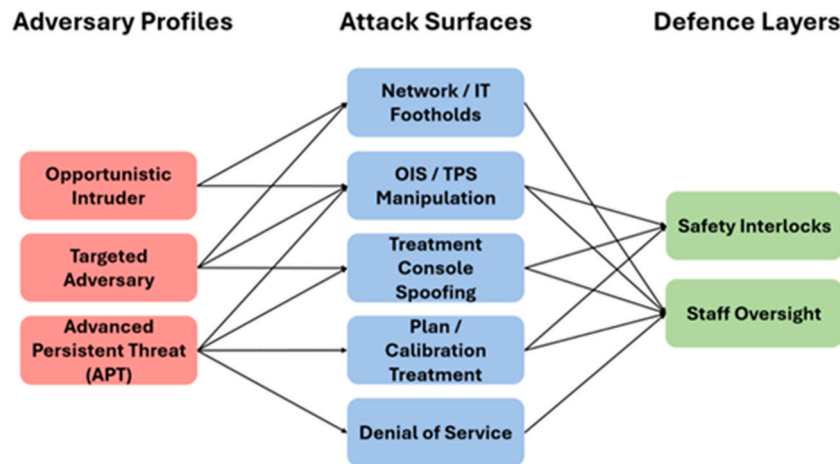


Fig. 2. Threat model for LINAC cybersecurity highlighting adversary profiles, attack surfaces and defence layers.

Such manipulations may remain difficult to detect if changes stay within clinically acceptable tolerance ranges and exploit the trust relationships between planning, verification, and delivery systems.

3. **Treatment console manipulation**, including spoofing of displayed parameters or suppression of alarms, which undermines the final point of human verification.
4. **Plan and calibration integrity**, where malicious changes to DICOM RT plans, couch coordinates, or beam calibration data could degrade treatment accuracy.
5. **Denial-of-service pathways**, where disruption of connected systems prevents or delays radiotherapy, potentially impacting time-sensitive cancer care.

As seen in Fig. 2, opportunistic intruders primarily exploit Network/IT footholds and, in some cases, accessible OIS/TPS components to disrupt availability or interfere with clinical workflows, whereas targeted adversaries with domain awareness are more likely to manipulate OIS/TPS data or treatment console interfaces to introduce subtle, workflow-consistent errors. APTs, while less likely, possess the capability to access a broader range of attack surfaces, including plan and calibration data, though such high-effort attacks are often unnecessary given the effectiveness of simpler upstream manipulations. The defence layers shown, including safety interlocks and staff oversight, operate predominantly at later stages of the radiotherapy workflow and therefore do not directly prevent upstream cyber intrusions. Instead, they serve as downstream safeguards that may limit the impact of certain attack paths or enable indirect detection during treatment setup and delivery. Connections in the figure indicate potential interaction or influence rather than guaranteed detection or uniform protection across all attack surfaces.

Vulnerabilities and attack vectors on LINACs

Altering patient alignment and positioning

Accurate patient positioning is critical in radiotherapy; even millimetre-scale errors can shift the high-dose region away from the tumour and into healthy tissue. Although patients are initially aligned using external skin marks and lasers, final positioning depends on digitally controlled and networked verification systems such as optical surface scanning and CBCT, which themselves represent potential attack surfaces. An attacker with network access could, for instance, spoof alignment feedback: in an optical surface monitoring system, falsified surface data might always report 'within tolerance,' misleading RTTs into believing the setup is correct. Similarly, couch positioning could be manipulated by altering digital coordinates or reference frames;

however, such deviations would only evade detection if tolerance thresholds or verification tables were themselves modified or bypassed, potentially allowing systematic positioning errors that place patients at risk. CBCT verification could also be compromised, either through manipulated registration calculations or falsified images that appear to match the planning CT. A sophisticated adversary could bypass safeguards by feeding false assurances and targeting human trust in guidance systems.

Likelihood assessment

Attacks targeting patient alignment and positioning systems are assessed as low likelihood. However, the potential clinical impact is high, particularly in high-precision treatments such as stereotactic radiosurgery, where small positioning errors can translate into clinically significant dose misdelivery. As a result, although technically non-trivial, this attack vector represents a credible concern under a targeted adversary model.

Spoofing the operator console and treatment parameters

Another high-impact vector is compromising the LINAC treatment console software itself. The console is where RTTs view and confirm the treatment settings (field size, gantry angle, MUs, etc.) before and during irradiation. If an attacker gained administrative access to the console computer, for instance through a phishing attack or a computer operating system vulnerability, they could deploy malware that alters the parameters sent to the LINAC in real time while the console still displays the physician-approved values. Such interface spoofing would be a form of man-in-the-middle (MITM) attack; it is plausible in principle because many consoles run on general-purpose operating systems. Malware could also suppress alarms that would normally warn of deviations, thereby undermining the console's role as the final point of human verification. While deep intrusion into the console is required, its network connectivity for R&V and remote support makes this pathway feasible. It should be noted, however, that not all treatment parameters can be modified solely through administrative access; depending on the vendor and system configuration, certain safety-critical settings may require additional hardware authentication, service modes, or physical dongles, thereby increasing the effort and reducing the practicality of such attacks.

Likelihood assessment

Console-level spoofing attacks are considered moderate likelihood relative to other vectors. Such attacks benefit from requiring no direct interaction with LINAC hardware and from exploiting the console's role as the final point of human verification, making this vector both feasible

and operationally attractive to a determined adversary.

Tampering with treatment plan data

Radiotherapy depends on a chain of data handoffs that, if compromised, could result in silent corruption of the treatment. One target is the treatment plan file itself. For some systems, treatment plans are exported from the TPS in DICOM RT format and imported into the ROIS, then retrieved by the LINAC console. Although many systems employ checksums, unique identifiers, or integrity verification mechanisms that make unnoticed plan modification difficult, these protections are not universally implemented or may be circumvented if integrity verification is compromised or disabled. Under such conditions, an attacker could intercept or replace plan files in transit or at rest and alter critical DICOM-RT parameters, potentially introducing clinically significant errors that may go unnoticed if downstream systems or console readouts are also compromised. This scenario is analogous to known exploits in diagnostic radiology. Mirsky et al. demonstrated 'CT-GAN' attacks in which AI was used to maliciously alter CT scans, adding or removing tumours in a way that radiologists were unable to detect [37,38]. Similar tampering in radiation oncology could target treatment prescriptions, calibration data, configuration files, or logs. These types of attacks mirror exploit in industrial control systems (ICS), where sensor and control data are falsified together to deceive operators and bypass detection mechanisms [39–42].

Likelihood assessment

Treatment plan tampering is assessed as a moderate likelihood attack vector. The feasibility of this attack increases when combined with console spoofing or workflow manipulation, as altered plans may be delivered without immediate detection. Given its scalability and indirect nature, plan tampering represents a realistic threat under both targeted and advanced adversary models.

Disabling safety interlocks and alerts

LINACs include multiple safety features to detect or halt abnormal behaviour, such as dose-monitoring ion chambers that terminate the beam if MUs exceed presets and software interlocks that stop treatment when parameters drift beyond tolerance. The ROIS and machine also generate logs and alerts for dose deviations and hardware faults. A cyber attacker might attempt to neutralise these safeguards by suppressing alarms or falsifying console messages. For example, malware on the console or control computer could filter warnings and consistently report normal status, delaying staff intervention. In practice, however, many safety interlocks are monitored by multiple, partially independent subsystems, which increases resilience against single-point failures. Consequently, more advanced attack scenarios would likely require interference with inter-subsystem communications between the controller, dose monitoring system, gantry and couch motion units, and imaging components, potentially delaying or blocking interlock signals. A more sophisticated approach could involve replaying previously recorded normal sensor data to mask hazardous conditions, as observed in replay attacks on industrial control systems [41,42]. Another potential vector is the compromise of automated QA or verification processes, leading to false pass results that could mislead clinical staff if such checks are relied upon in isolation.

Likelihood assessment

Direct disabling of hardware safety interlocks is considered low likelihood. However, indirect interference, such as suppressing alerts, falsifying system status indicators, or replaying normal sensor data, presents a low to moderate likelihood pathway when combined with prior system compromise. Although technically demanding, such approaches have been demonstrated in other industrial control environments and could be attractive to highly capable adversaries seeking

stealth rather than overt system failure.

LINAC layered attack escalation perspective

As healthcare and industrial control networks converge, scenarios once considered theoretical are becoming increasingly plausible, individually or in combination. The overarching objective of a stealth-oriented attacker would be to make the system appear to operate within normal limits while treatment delivery is compromised. Fig. 3 presents a conceptual layered attack escalation perspective for LINAC environments. The figure illustrates a progression of increasingly capable attack classes, moving from network-level footholds to socio-technical and work-flow level manipulation. The layers represent escalation in attacker capability and system impact, not mutually exclusive or sequential steps. The potential cyberattack pathways in a LINAC environment, can be viewed in layers:

1. **Network-level intrusions**, where the attacker gains a foothold in the hospital or radiotherapy network (e.g., through phishing, unsecured remote access, or an infected device). This serves as the entry point for subsequent actions.
2. **Software-level exploits**, where ROIS or TPS systems, which may run on standard operating systems, are compromised, allowing manipulation of treatment data or denial of service.
3. **Device firmware or controller exploits**, where the attacker directly targets the LINAC's control firmware or calibration settings. This is the most difficult but also the most powerful level, as it could override even local safety checks.
4. **MITM and interface exploits**, where communication between systems (TPS–ROIS–LINAC or within LINAC subsystems) is intercepted and altered, enabling stealth modifications of parameters.
5. **Socio-technical and workflow-level exploits**, where attackers exploit clinical workflows, trust assumptions, or automated quality-assurance processes rather than directly intercepting communications. Examples include falsified QA logs, manipulated audit trails, misleading system status indicators, or altered alert thresholds that cause operators or physicists to rely on compromised outputs. Unlike

LAYERED ATTACK ESCALATION PERSPECTIVE

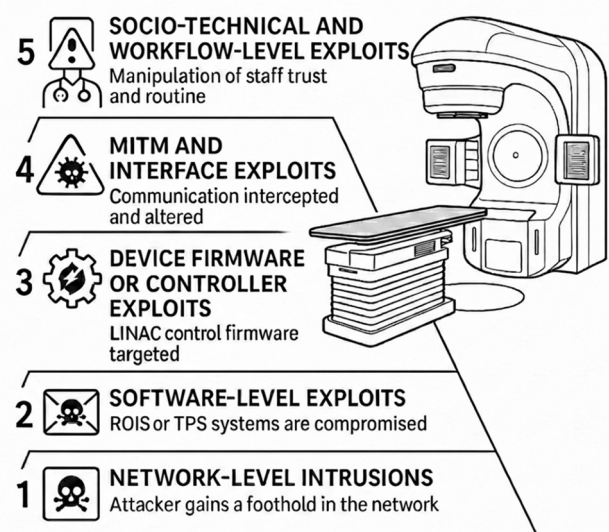


Fig. 3. Layered attack escalation perspective for LINAC environments, illustrating increasing attacker capability and impact. The visual hierarchy is used for clarity and should not be interpreted as a strict attack dependency graph or probability ranking.

interface-level exploits, these attacks operate at the level of system use and validation rather than data transmission.

Defence mechanisms and cybersecurity solutions

The defence mechanisms discussed in this section are framed to reflect the operational workflow of radiation oncology departments. Not all measures are expected to be implemented or managed by clinical staff. Instead, responsibilities are distributed across stakeholders: LINAC vendors are primarily responsible for the design, certification, and secure implementation of device-level and firmware-level protections; hospital IT and clinical staff oversee secure deployment, configuration, and monitoring within the clinical environment; and frontline clinical users interact with these systems through established workflows that prioritise patient safety, efficiency, and time-critical treatment delivery. Throughout this section, recommended controls are therefore discussed with consideration of clinical workflow constraints, regulatory requirements, and the division of responsibility between vendors and healthcare operators.

Risk assessment and network segmentation

The first step in securing a LINAC environment is a thorough risk analysis of the radiation oncology network. In this context, network segmentation refers to dividing the clinical network into isolated zones that restrict which systems can communicate with each other, firewalls enforce policy-based filtering of allowed traffic between these zones, virtual local area networks (VLANs) logically separate devices at the network layer, and demilitarized zones (DMZs) provide controlled gateways for external or vendor access without exposing internal clinical systems. Hospitals should conduct regular vulnerability assessments to identify insecure configurations, outdated software, and potential entry points. The LINAC control network (OT domain) should be segmented from the general hospital IT network using VLANs, firewalls, based segmentation, and strictly controlled communication interfaces [43,44]. In practice, complete physical isolation of LINAC systems is uncommon due to clinical and operational requirements for imaging, treatment planning, verification, and vendor support; consequently, security is typically achieved through defence-in-depth and tightly controlled network segmentation rather than full disconnection. For instance, the treatment console and LINAC controller may reside on a dedicated subnet that communicates with the ROIS only through tightly controlled interfaces without direct internet access. Further micro-segmentation can isolate the TPS, ROIS database server, and machine controllers into separate zones, limiting lateral movement if one system is compromised [45]. Segmentation should also apply to remote support pathways [45], meaning that vendor remote access must be logically isolated from both hospital IT and radiotherapy OT networks, restricted to predefined systems and services, and enabled only through tightly controlled DMZs or VPN gateways with strict authentication, authorization, and time-limited access. Vendor documentation reinforces this principle: Varian's Halcyon Product Planning Guide specifies VLANs and firewall rules to isolate treatment delivery [46], while Elekta's Network Security Solution recommends defence-in-depth with segmentation as the first containment measure [47].

Implementing cybersecurity frameworks and best practices

Healthcare institutions should adopt established frameworks to strengthen defences. The NIST Cybersecurity Framework (CSF) defines five core functions (Identify, Protect, Detect, Respond, and Recover) which in radiation oncology translate to asset identification, access control, anomaly detection, incident response, and recovery through backups and contingency workflows [48,49]. ISA/IEC 62443, an international ICS security standard for medical OT systems, provides component-, system-, and policy-level controls, supporting secure

development and integration [50]. Radiation oncology teams should also maintain contingency plans for cyber incidents [23,27]. Experience from real attacks highlights the need for predefined workflows, such as backup dose schedules and printed patient data [6,19]. Regular drills should rehearse response procedures, while technical hardening (removing unnecessary software, closing unused ports, and applying vendor-recommended patches) remains essential. While such frameworks may not directly prevent or immediately detect stealth manipulations of treatment data or interfaces, they play a critical role in reducing attacker dwell time, enabling retrospective detection through audit trails and anomaly investigation, and supporting controlled recovery once anomalous behaviour is suspected or uncovered.

Access control and zero-trust architecture

Strong authentication and authorization are vital to prevent unauthorized LINAC manipulation. All ROIS, TPS, and console accounts should use unique passwords, and default vendor credentials must be changed [46,47]. Where technically feasible, multi-factor authentication (MFA) should also be implemented. However, MFA is not universally supported across all radiotherapy systems and may require coordination between vendors and hospital IT infrastructure before it can be enabled. While MFA is often assumed to require a secondary external device (e.g. mobile phones), this is not the only implementation model. In clinical environments, MFA can be realised through integrated and non-disruptive mechanisms, such as biometric authentication (e.g. fingerprint, facial, or iris recognition), hardware-backed credentials, or context-aware session authentication, which do not require additional user devices and can be compatible with high-frequency clinical workflows. Access should be limited to authorized staff, while portable media (e.g. USB drives) must be restricted to reduce malware risks [44,45]. A zero-trust model ensures no user or device is implicitly trusted. TPS workstations should only accept authenticated connections, and role-based access control (RBAC) should enforce privileges. All activities must be logged, stored in tamper-resistant or immutable systems, and regularly audited. Intrusion detection systems (IDS) tailored for medical networks can further support anomaly detection [44,45]. For maintenance and service access, stronger identity and access management (IAM) approaches are recommended to ensure that elevated privileges are scoped, time-limited, and auditable rather than permanently enabled. For example, recent industrial security proof-of-concepts (NAMUR/ZVEI) aligned with NE201 (a guideline for secure, identity-based access control in industrial and operational technology systems) demonstrate the use of short-lived, role-scoped JSON Web Tokens (JWTs) bound to proof-of-possession (PoP) keys, which cryptographically verify the identity, role, device, and session context of a user before access is granted [51]. Applied to LINACs, such mechanisms could control service modes, configuration changes, calibration edits, and remote-assist sessions while preserving treatment delivery interlocks.

Importantly, access control mechanisms in radiotherapy environments must be designed to respect clinical workflow constraints and time-critical treatment delivery. High-frequency actions performed by RTTs during treatment sessions cannot rely on repetitive or disruptive authentication steps. For this reason, the access control mechanisms discussed above are intended to be role-aware, context-sensitive, and selectively enforced, rather than uniformly applied at every interaction. In practice, stronger authentication and session verification can be concentrated at workflow boundaries (e.g. session initiation, plan approval, calibration changes, or remote service access), while routine treatment execution remains streamlined. This approach aligns security controls with established clinical roles and responsibilities, ensuring patient safety and operational continuity without imposing undue cognitive or time burden on clinical staff.

Intrusion detection and anomaly monitoring

Because stealthy attacks may evade standard safeguards, advanced monitoring is essential. Network-based intrusion detection and prevention systems (IDS/IPS) can flag anomalies such as unfamiliar devices or abnormal command sequences on LINAC control ports [44]. Device-level monitoring of operational data (e.g., dose rates, interlock statuses) can be enhanced with analytics or AI to detect subtle deviations, such as beam output shifts that still pass daily QA. Two-layer anomaly detection has been proposed: the first verifies that high-level parameters match the treatment plan, while the second analyses device behavior for physical inconsistencies [37]. Real-time log streaming could expose gradual manipulations, such as incremental MU increases leading to overdose. Independent safeguards include redundant dose verification via a secondary MU calculator or external dose watchdog. If discrepancies are found, these isolated systems can trigger alerts. In addition, RTTs and physicists must be trained to identify abnormal conditions, such as frozen interfaces or contradictory displays, and respond with immediate safety protocols, including halting treatment and switching to standby systems.

Cryptographic protections and data integrity

Maintaining treatment data integrity is essential. Digital signatures or cryptographic checksums can secure plan files and parameters, allowing the ROIS and LINAC console to reject altered data [44,45]. Similarly, cryptographic authentication of command messages prevents man-in-the-middle attacks. While modern LINACs increasingly support encryption and authentication [46], older models using plain text protocols should be updated or encapsulated within secure channels such as VPNs. Data at rest in the ROIS must also be encrypted and backed up offline to mitigate ransomware risks, while configuration and calibration data should be stored in secure, verifiable baselines for recovery [44,45,52]. At the device level, elliptic curve cryptography (ECC), such as the widely used secp256r1 curve, are computationally feasible even on low-power embedded controllers. For example, on a representative microcontroller platform (STM32WBA55, 32 MHz), Elliptic Curve Digital Signature Algorithm (ECDSA-P256) signature verification completes in approximately tens of milliseconds with minimal power consumption, demonstrating that cryptographic integrity checks can be performed without impacting real-time control [51]. For LINAC systems, this implies that digital signatures and proof-of-possession mechanisms can be used to verify the authenticity of firmware updates, configuration changes, or privileged commands before execution. Trust anchors, such as identity provider (IdP) public keys, can be securely stored on the device to validate signed tokens containing device-specific attributes and expiration times. In environments with restricted connectivity, token validity can be managed using predefined time windows, persistent credentials, or controlled break-glass procedures to maintain operational continuity while preserving cryptographic assurance [51].

Vendor and manufacturer security measures and Shared operational responsibilities

LINAC manufacturers play a central role in cybersecurity by designing, validating, and maintaining certified security mechanisms within LINAC systems [53,55]. Hospitals, in turn, are responsible for operating these systems securely, working closely with vendors to apply approved updates, manage access, and ensure staff are trained on built-in safeguards [45]. For example, Elekta's Network Security Solution hardens connections between LINACs and external networks [47]. In practice, security responsibilities in radiation oncology environments are shared but distinct. Vendors are responsible for the design, certification, and secure implementation of device-level controls, while local hospital administrators are responsible for secure operation, oversight, and policy enforcement within the clinical environment. Such tools

should be configured and maintained primarily by the vendor, in accordance with regulatory and certification requirements, and performed under the supervision and approval of authorised local hospital administrators. This includes changing default credentials, enabling firewalls, disabling unnecessary Windows services, enforcing application whitelisting, and preventing console PCs from being used for general browsing or email [54]. Remote access should be restricted to supervised vendor service sessions, initiated and controlled by the vendor, with local administrators approving access windows and, in some centres, enabling connectivity only during scheduled maintenance. Hospitals should also maintain an inventory of software versions and track disclosed vulnerabilities. Regulatory bodies such as the FDA now require cybersecurity to be integrated into device design and post-market vulnerability management [44,45,52,55]. Compliance with these standards and rapid patching are essential, while open vendor communication can expedite fixes for zero-day exploits. The following measures primarily concern vendor-side system design and service integration, rather than day-to-day hospital administration. Emerging best practices for vendor integration include [51]:

- Using IdP and public key infrastructure (PKI) with short-lived, role-based access tokens (e.g. AD/LDAP-backed) to ensure that vendor service access is time-limited, auditable, and restricted to authorised roles during maintenance activities.
- Shipping consoles and controllers with embedded trust anchors and token validators (e.g. JWT and ECC) to cryptographically verify that privileged service commands originate from authenticated vendor tools and authorised service sessions, enforcing proof-of-possession before execution.
- Standardising token formats and provisioning workflows to enable consistent auditing, revocation, and cross-site portability of service credentials across multi-vendor clinical environments.
- Supporting isolated or connectivity-constrained deployments through mechanisms such as Online Certificate Status Protocol (OCSP) stapling, short-lived signing keys, and secure offline time validation, allowing cryptographic assurances to remain effective even when continuous external connectivity is not available.

Resilience and recovery planning

Even with strong prevention, departments must prepare for cyberattacks that disrupt LINAC operations. Resilience requires both technical and procedural safeguards. Technically, offline backups of treatment plans, machine profiles, and patient data are essential [45]. If the ROIS is compromised; fallback options include printed plans or simplified treatments delivered directly via the console [27]. Some centers also maintain read-only secondary ROIS servers or cloud-based backups for rapid activation [6]. Procedurally, staff should be trained in contingency workflows, including stand-alone LINAC operation, manual dose calculations, and rescheduling treatments after restoration. Lessons from the 2021 Irish HSE cyberattack highlight crisis measures such as patient prioritisation, regulator communication, and dose compensation for interruptions [11,13]. Embedding these practices into departmental protocols ensures that even if attacks occur, patient harm is prevented through robust fallback measures. While the measures described in this section do not directly prevent or immediately detect stealth manipulations of treatment parameters, they play an important complementary role in mitigating their consequences and enabling post-incident discovery. Offline backups, immutable records, and independent treatment plan archives provide trusted reference points that can be used to retrospectively identify discrepancies once anomalous outcomes, QA deviations, or clinical concerns arise. Similarly, contingency workflows, such as stand-alone LINAC operation, manual dose calculations, and simplified fallback treatments, limit patient harm by reducing reliance on potentially compromised digital systems during recovery. In this sense, resilience and recovery mechanisms do not replace preventive or

detective controls but instead bound the clinical impact of stealth attacks and support forensic validation and safe continuation of care following a suspected compromise.

Emerging protective technologies

Several emerging approaches, largely requiring manufacturer-level integration or next-generation system designs, could further strengthen LINAC security against stealthy manipulation. Behavioral whitelisting for OT devices, implemented at the controller or firmware level by manufacturers, ensures only pre-approved commands are executed, blocking abnormal inputs [56]. Machine learning (ML) techniques, when embedded into vendor software pipelines or planning systems, can detect subtle deviations in treatment patterns, including anomalies in DICOM images [57]; autoencoder-based methods have already been applied to lung cancer VMAT plans [58]. Digital twin models, which compare real-time LINAC behavior against a virtual replica, may help identify tampering [59]. From a hardware perspective, future LINAC designs could integrate redundant safety controllers that cross-verify outputs, making compromise more difficult, and incorporate hardware protections such as trusted platform modules (TPMs) to ensure that only authenticated firmware is executed.

Human and organizational factors

Policy and personnel play a critical role in cybersecurity. Regular staff training alone is unlikely to detect stealth manipulation of treatment systems, though, education on recognising phishing attempts, reporting anomalous system behaviour, and adhering to secure operational practices strengthens baseline cyber hygiene and reduces the likelihood of initial compromise. Together with technical safeguards, these measures contribute to a layered defence that lowers the overall risk of successful stealth attacks. The culture in radiation oncology should treat cybersecurity with the same priority as patient safety. Institutional policies must clearly define responsibilities during a cyber incident, including coordination with IT, communication with vendors, and informing patients of treatment delays. Cybersecurity drills, similar to emergency code drills, can reinforce this culture. Beyond the clinic, stronger industry-wide standards are also needed. Professional organisations such as the American Association of Physicists in Medicine (AAPM) and the American Society for Radiation Oncology (ASTRO) have begun addressing cybersecurity [60,61], and accreditation bodies are expected to adopt requirements in the near future.

Summary of defence mechanisms

Fig. 4 summarises nine essential cybersecurity defence mechanisms for protecting LINACs in radiation oncology. Effective defence requires a layered approach: securing network pathways, hardening and monitoring devices, enforcing strict access controls, maintaining robust backup and recovery processes, and fostering a security-aware clinical culture. No single safeguard is sufficient; resilience arises from the integration of multiple technological and human measures. Core practices such as cryptographically signed treatment plans, zero-trust network architecture, rigorous multi-step verification, and segmented networks are not optional but essential to counter stealthy cyber threats in radiation oncology. These measures emphasise that effective LINAC cybersecurity is not achieved through isolated technical controls, but through coordinated vendor design, secure clinical operation, and workflow-aware deployment that preserves treatment efficiency while reducing the risk and impact of stealth attacks.

To clarify the practical implementation of these measures, Table 1 summarises each defence mechanism according to its primary responsibility and implementation feasibility within current clinical practice. The table highlights which controls can be realistically deployed at the hospital or departmental level, which depend on vendor-

Defence Mechanisms



Fig. 4. Summary of key cybersecurity defence mechanisms for protecting LINACs in Radiation Oncology.

side design and certification, and which represent emerging or future capabilities. This distinction reflects real-world clinical workflow constraints, regulatory requirements, and the shared responsibility between manufacturers and healthcare operators.

Limitations & future work

While this review highlights pathways for LINAC stealth cyber manipulations and surveys a broad range of defensive strategies, several limitations must be acknowledged. First, the feasibility of many attack vectors, particularly those involving manipulation of firmware, safety interlocks, or low-level control logic, is inherently dependent on vendor-specific system architectures, redundancy designs, and certification constraints. Modern LINAC platforms incorporate multiple layers of safety checks and independent subsystems, and the extent to which these mechanisms can be bypassed varies significantly across manufacturers and device generations. A systematic, device-specific evaluation of which classes of attacks are realistically achievable, and under what assumptions, requires close collaboration with equipment manufacturers and access to proprietary design details, and therefore falls outside the scope of this review. Second, although the paper discusses defence mechanisms spanning network security, access control, cryptographic integrity, monitoring, and resilience, not all proposed measures are equally deployable in current clinical settings. Practical implementation is constrained by regulatory approval processes, certification timelines, staffing availability, clinical workload, and allowable system downtime. As clarified throughout the *Defence Mechanisms and Cybersecurity Solutions* section, several controls, particularly those at the firmware, controller, or hardware level, can only be implemented by vendors as part of certified system updates, while hospitals and clinics retain responsibility for secure configuration, access governance, monitoring, and operational procedures. Third, this work focuses on

Table 1
Practical feasibility and responsibility of cybersecurity defence mechanisms.

Defence mechanism	Primary responsibility	Practical feasibility today	Clinical workflow, regulatory, and operational considerations
Risk assessment and network segmentation (VLANs, firewalls, DMZs)	Hospital IT, with vendor guidance	High	Widely deployable within current clinical environments; security is typically achieved through defence-in-depth rather than full physical isolation, preserving required imaging, planning, verification, and vendor support workflows
Regular vulnerability assessment and system hardening	Hospital IT/clinical administration	High	Periodic process with no real-time impact on treatment delivery; requires coordination with vendors to remain within supported configurations
Implementation of cybersecurity frameworks (NIST CSF, IEC 62443)	Hospital IT/institutional leadership	High	Provides organisational structure across Identify–Protect–Detect–Respond–Recover functions; does not directly prevent stealth manipulation but reduces attacker dwell time and supports controlled recovery
Contingency planning, backups, and predefined fallback workflows	Clinical leadership/department	High	Essential for maintaining patient care during cyber incidents; includes printed plans, manual calculations, stand-alone LINAC operation, and treatment prioritisation
Access control, role-based access, and selective multi-factor authentication	Shared (vendor and hospital)	Moderate	Stronger controls applied at workflow boundaries (e.g. plan approval, configuration changes, service access), while routine RTT actions during treatment remain streamlined
Zero-trust architecture and authenticated system interactions	Hospital IT, with vendor compatibility	Moderate	Requires careful integration to avoid disruptive or repetitive authentication in time-critical clinical workflows
Intrusion detection and network-based anomaly monitoring	Hospital IT	Moderate	Passive monitoring is largely workflow-neutral; effectiveness depends on OT-aware configuration and alert interpretation
Device-level and treatment-data anomaly monitoring	Shared, primarily vendor-side	Low to Moderate	May detect subtle deviations that pass routine QA; typically retrospective and not yet a primary real-time safeguard
Cryptographic protection of treatment plans, commands, and data integrity	Vendor	Moderate (modern systems)	Increasingly supported on newer platforms; retrofitting legacy systems may be limited by certification and regulatory constraints
Firmware-level security, trust anchors, and proof-of-possession mechanisms	Vendor	Low (current); high (future)	Requires manufacturer-level design and regulatory validation; minimal clinical workflow impact once implemented
Secure vendor remote access and service session control	Vendor, with hospital oversight	Moderate	Access is time-limited, auditable, and restricted to authorised service sessions, preserving clinical safety and accountability
Resilience and recovery mechanisms following cyber incidents	Shared (vendor and hospital)	High	Offline backups, immutable records, and independent reference data limit patient harm and support post-incident validation
Emerging protections (ML-based detection, digital twins, hardware security)	Vendor/research	Low (emerging)	Promising approaches requiring next-generation system design and regulatory acceptance; not routine clinical practice
Human and organisational measures and cybersecurity culture	Institution/department	High	Education and policy reduce the likelihood of initial compromise but cannot independently detect stealth manipulation

conceptual threat modelling and risk pathways, rather than empirical exploitation or system testing. We intentionally avoided presenting proof-of-concept attacks on clinical LINAC systems, as such testing would require controlled laboratory environments, ethical approval, and vendor cooperation to ensure patient safety. Future work should prioritise validated testbeds, digital twins, or manufacturer-supported evaluation platforms to experimentally assess stealth attack feasibility and the effectiveness of proposed mitigations under realistic operating conditions. Finally, future research should explore quantitative risk assessment, including the relative likelihood and impact of different attack classes across clinical contexts, and the development of workflow-aware security mechanisms that better align cyber protections with time-critical radiotherapy delivery. Greater integration between cybersecurity engineering, clinical physics, and vendor design processes will be essential to translate these insights into deployable, safety-certified solutions.

Conclusions

The digitalisation of radiation oncology has brought unprecedented opportunities for precision and efficiency, but it has also created new vulnerabilities. Although no confirmed cases of malicious cyberattacks on LINACs have been reported, history from other critical sectors shows that absence of evidence is not evidence of absence. As healthcare and industrial control systems converge, threats once viewed as theoretical are becoming increasingly plausible. Because LINACs operate within interconnected workflows involving the TPS, ROIS, and treatment console, their safe function depends on the integrity of the entire ecosystem. This review has highlighted current cybersecurity challenges, moving beyond the well-documented ransomware incidents that disrupt care to consider the risk of other cyberattacks and stealth manipulations. Strong interlocks and vigilant human oversight make catastrophic failures

unlikely, yet radiation oncology departments have already experienced IT outages and ransomware events with direct impact on patient care. Subtle upstream manipulations, while less probable, remain a credible concern and should be recognised as plausible under certain conditions.

Mitigating these risks requires a proactive, multi-layered approach. Cybersecurity for LINACs should be treated as a continuous process, starting with secure-by-design principles, regular vulnerability assessments, and controlled penetration testing where appropriate. Regulatory agencies such as the FDA and EU bodies, along with professional societies, are beginning to issue relevant guidance, but more detailed frameworks tailored to radiation oncology are still needed. Progress will depend on close collaboration among clinical staff, IT and security teams, vendors, and regulators. Training, awareness, and a culture of prompt reporting are as essential as technical safeguards. Finally, vendors play a central role in defining which classes of attacks are realistically feasible and at which system layers mitigation is most effective. Vendor-led guidance, certified security updates, and transparent post-market vulnerability management are therefore essential components of future risk reduction. These precautions are not meant to discourage radiotherapy; the benefits of timely treatment overwhelmingly outweigh speculative risks. Rather, strengthening LINAC cybersecurity is an investment in both patient safety and the resilience of cancer care in the digital age.

Author contributions

A. A. & M.A.: conceptualized the article and developed its structure. A.A. and M.A.: wrote the main content of the review article. N. C., A. M. M., and F. B.: contributed revisions, feedback, and suggestions to enhance the article. All authors read and agreed to the published version of the manuscript.

CRediT authorship contribution statement

Abdulaziz Alhussan: Writing – review & editing, Writing – original draft, Investigation, Conceptualization. **Mohammad Alhussan:** Writing – review & editing, Writing – original draft, Investigation, Conceptualization. **Nick Chng:** Writing – review & editing. **Francesca Boem:** Writing – review & editing, Supervision. **Anna Maria Mandalari:** Writing – review & editing, Supervision.

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